

Learning Multi-View Interactional Skeleton Graph for Action Recognition

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Abstract—Capturing the interactions of human articulations lies in the center of skeleton-based action recognition. Recent graph-based methods are inherently limited in the weak spatial context modeling capability due to fixed interaction pattern and inflexible shared weights of GCN. To address above problems, we propose the multi-view interactional graph network (MV-IGNet) which can construct, learn and infer multi-level spatial skeleton context, including view-level (global), group-level, joint-level (local) context, in a unified way. MV-IGNet leverages *different skeleton topologies* as multi-views to cooperatively generate complementary action features. For each view, separable parametric graph convolution (SPG-Conv) enables multiple parameterized graphs to enrich local interaction patterns, which provides strong graph-adaption ability to handle irregular skeleton topologies. We also partition the skeleton into several groups and then the higher-level group contexts including inter-group and intra-group, are hierarchically captured by above SPG-Conv layers. A simple yet effective global context adaption (GCA) module facilitates representative feature extraction by learning the input-dependent skeleton topologies. Compared to the mainstream works, MV-IGNet can be readily implemented while with smaller model size and faster inference. Experimental results show the proposed MV-IGNet achieves impressive performance on large-scale benchmarks: NTU-RGB+D and NTU-RGB+D 120.

Index Terms—Action recognition, skeleton, multi-view, graph neural network, hierarchical method

1 INTRODUCTION

ACTION understanding lies in the center of human-related video analysis, e.g., visual surveillance, autonomous driving, and robotic interactions, *etc.* Various modalities have been exploited to address action analysis problem, including visual appearance (RGB), depth information (RGB-D), optical flow, even sound. RGB image-based action understanding (e.g., action recognition, prediction, and detection) has made a breakthrough by taking advantages of deep convolution neural network (CNN). Abundant works [1], [2], [3], [4] have proved that fine-grained motion dynamic is a critical clue to discriminate different actions. However, directly extracting action representations from video sequences neglects essential fine-grained features, especially the intrinsic interactions between human articulations.

Compared to the RGB image, 2D/3D human skeleton can be generated by pose estimation methods (e.g., Openpose [5], Densepose [6]), or captured by Microsoft Kinetics [7], showing its irreplaceable advantages in action representation. Namely, it provides detailed position and movement information of human articulations (joints in skeleton), which are essential for constructing spatiotemporal and motion

features. Although many works [8], [9], [10] have utilized skeleton data to help human action understanding, e.g., skeleton-based action recognition, how to efficiently capture spatial interaction context (the interactions among human articulations), is still a key challenge.

For skeleton-based action recognition, recent works mainly learn action representations with deep neural networks (DNN). There are generally three ways: the first type [11], [12] utilizes recurrent neural networks (RNN), e.g., Long-Short Term Memory (LSTM), to model sequential feature vectors which are formed by concatenating joint coordinates at each frame. This type learns temporal dynamics of feature vectors without considering spatial context, i.e., it generates spatial features directly through fully-connection (FC) and ignores the relationships of skeleton joints. The second [13], [14] regards the spatial and temporal dimensions of skeleton sequence as image's width and height respectively to form spatiotemporal skeleton maps and performs vanilla 2D convolution on above maps to capture spatiotemporal co-occurrence features. The essential spatial relationships are still ignored for this type. The third way constructs skeleton graph according to *physical* connections of human articulations [8], [9], [15], [16], [17], and extracts spatial features by graph-based methods, e.g., graph convolution network (GCN).

Although GCN-based methods improve the capabilities of capturing the articulations' relationships to some degree, fixed interaction pattern and inflexible shared weights of GCN still limit its expressiveness. Notice that discriminating complex actions needs to capture the abundant and dynamic relationships of nodes/joints in the skeleton sequence. However, in previous graph-based works, *fixed graph* means constant interaction pattern among joints (static rather than dynamic), and *weight sharing* leads to a single interaction pattern (poor instead of abundant), which

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Manuscript received 29 September 2019; revised 30 September 2020; accepted 10 October 2020. Date of publication 21 October 2020; date of current version 5 May 2023.

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Recommended for acceptance by S. Bai, P.H.S. Torr, R. Krishna, F.-F. Li, A. Gupta, and S.-C. Zhu.

Digital Object Identifier no. 10.1109/TPAMI.2020.3032738

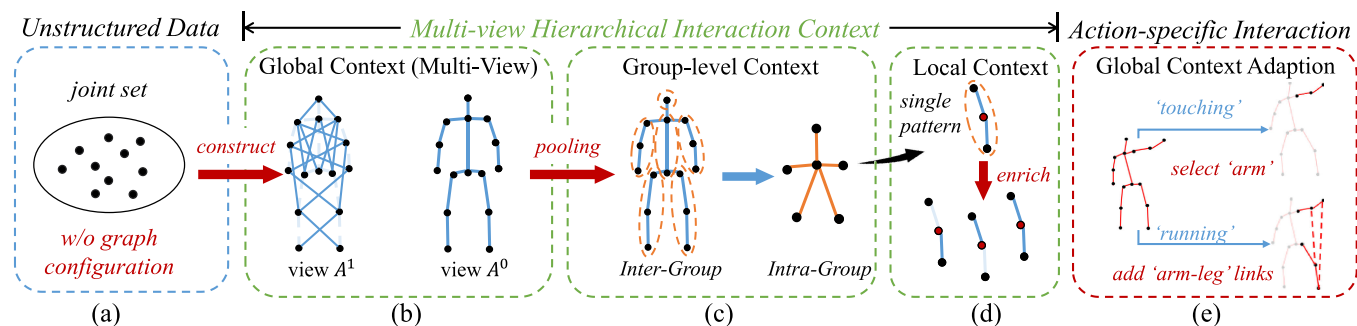


Fig. 1. The multi-level skeleton context represented by hierarchical graphs. (a) Original spatial skeleton data without any graph configurations. (b) Global graph/view-level context defines the “view” that we describe the skeleton. Fixed graph restricts the information flow path of joints. Our MV-IGNet aims to generate different graphs (multi-view) without common edges to extract complementary features. (c) The group-level context in each view, including intra-group and inter-group context, which possesses high-level semantic information. (d) Local context (interactions between a joint and its neighbors) is the cornerstone of skeleton feature extraction. We enable multiple learnable graphs to enrich local interaction patterns. (e) Action-related context. We expect that different interactions contribute unequally to different actions.

cannot well address the intrinsic multi-level spatial skeleton context shown as follows.

Specifically, there are multi-level spatial skeleton relationships in Fig. 1, in which each level context contains unique information. Joint-level (local) context describes the relationships between one joint and its neighbors. The group-level context shows joints’ latent relations and the higher-level interactions between groups. Graph/View-level (global) context defines all the possible interaction path of joints, i.e., the skeleton topology, which is very vital for describing and understanding actions.

Joint-Level Context. Existing GCN-based methods conduct weight-sharing mechanism for graph-structured feature extraction, which is indeed practical in handling dynamics graphs with arbitrary adjacent relations. But for skeleton data, single and shared weight means that for each joint, its neighbors always provide constant contributions, i.e., single interaction pattern. As shown in Fig. 1d, to thoroughly represent complex actions, the relationships between a joint and its neighbors (local context) should be more diverse. Thus, employing multiple learnable graphs can enrich the local interaction patterns and offer stronger feature extraction ability.

Group-Level Context. Current graph-based methods directly integrate features of all joints to generate final spatial features, which ignores the multi-level interactions. In both CNN and GCN, hierarchically capturing local context can facilitate high-level feature extraction, e.g., *pooling*. Thus, the hierarchical skeleton context modeling can also generate more comprehensive features for action recognition (See Fig. 1c). For instance, we can cluster joints in “right arm” as a group, and then the complex action “running” can be represented as the relationship between groups, i.e., the interactions between “arms” and “legs”.

Global-Level Context. The skeleton topology is fixed and manually defined according to human intuition in previous works, e.g., *physical* connections of articulations. This hand-craft approach indeed brings appreciable improvements but ignores other significant interactions, e.g., the relationships between the “left hand” and “right hand”. Though intuitively adding auxiliary connections can consider above ignored significant interactions, e.g., connecting “legs” and “arms”, these manual solutions are still inflexible and inefficient when dealing with different actions. Note that different

joints usually possess unequal importance for recognizing specific actions (See Fig. 1e), e.g., the action “touching” is mostly related to the joints in “hands” instead of “legs”. This characteristic motivates us to design a framework which can adaptively learn the skeleton topology rather than defining their interaction patterns.

Multi-View Representation. The skeleton graph can be regarded as a way for both human and neural network to describe and understand actions. We have known that different camera views can provide complementary information about the skeleton, which is usually represented as affine transformations on original coordinates. Intuitively, different skeleton topologies can also be regarded as transformations on the relationships of joints, which is also a critical perspective to offer complementary information (See Fig. 1b). Similar to the “isomers” in chemistry, two skeleton graphs that have the same joints but different topologies may reflect distinct properties of the human skeleton, i.e., ignoring and emphasizing different interactions. Therefore, we can discover two different skeleton topologies to cooperatively represent the same action for more robust recognition. However, except *physical* connections, existing graph-based methods are very sensitive to other graphs and cannot easily handle irregular topologies except tailor-made one. To this end, we expect to develop a simple, flexible and efficient framework that can hugely lower the requirements of network for the skeleton graph.

Inspired by above observations, we propose Multi-View Interactional Graph Networks (MV-IGNet) focusing on joints’ relationships, to construct, learn, and infer the multi-level spatial skeleton context in a hierarchical way for action recognition. MV-IGNet leverages a skeleton graph and its complement graph to cooperatively generate complementary action features. For each graph, Separable Parametric Graph Convolution (SPG-Conv) is devised to efficiently capture richer local context, which possesses strong adaption ability to handle irregular skeleton topologies. This superiority of SPG-Conv lies in enabling multiple learnable graphs as parameters to learn optimal interaction patterns. We also construct and extract higher-level group context based on hierarchical SPG-Conv layers. Global Context Adaption (GCA) module is also proposed to adaptively learn input-dependent skeleton topologies in the inference process. Different from the self-attention mechanism, GCA directly

operates on the weights of SPG-Conv rather than input features. In this way, the spatial interactional context including view-level, group-level, joint-level, as well as the input-related context is explicitly captured by MV-IGNet in a unified way as shown in Fig. 1.

Our main contributions are summarized below:

- A novel multi-view solution for action recognition, which cooperatively utilizes different skeleton topologies instead of camera views to obtain complementary action features. It's general for graph-based methods.
- A radical unified hierarchical modeling framework to capture the multi-level spatial skeleton features.
- A simple yet efficient SPG-Conv, different from GCN, that leverages multiple parametric graphs to learn relationships and weights simultaneously. It owns strong feature-extraction and graph-adaption capability while hugely reducing computational cost.
- An GCA module to "select" proper input-related interactions by directly operating on SPG-Conv weights instead of input features.
- Experimental results show the proposed MV-IGNet achieves impressive performance on large scale datasets: NTU-RGB+D and new NTU-RGB+D 120.

2 RELATED WORK

Graph-Based Neural Network. Graph neural network owns strong abilities of modeling and irregular domain. Graph based models usually include recurrent neural networks (RNN) [18], [19] and convolutional neural networks (CNN) [20]. For RNN based graph models, previous works [21], [22] formed a graph then apply RNNs to capture node dependencies. For graph convolution network (GCN), one typical implementation lies in the spectral perspective, considering graph convolution locality to perform spectral analysis [20], [23], [24]. Another typical fashion focuses on the spatial domain, where the convolution is directly applied to the graph nodes and their neighbors [25], [26]. Our work devises a novel graph-based framework, MV-IGNet, to capture multi-level spatial skeleton features.

Multi-View Learning. Multi-view learning is an emerging direction in machine learning which considers learning with multiple views to improve the generalization performance. Related works in machine learning can be found in recent surveys [27], [28]. Multi-views have also been widely used in many computer vision tasks. Human and objects can be described from different points of view (camera views) [29]. Color/shape and texture of objects are also two different kinds of features, which can be regarded as two-view data [30]. In many cases, however, no natural multiple views are available due to certain limitations, so that only one view may be provided to represent the data. Some works have attempted to create more views from single view by specific data augmentation, e.g., style transfer [31] or by complementary constraints [32], [33]. In this paper, different from prevalent camera views in action recognition, we leverage different skeleton topologies to create multi-views. Different graphs provide distinct ways to describe and understand the skeleton for both human and machines, which indeed can offer complementary information.

Self-Attention Mechanism. Self-attention [34] has made a great progress in machine translation. Recently, many applications of self-attention have been introduced into various computer vision tasks. Their similarities lie in computing the response at a position (e.g., in the sentence, image, or video sequence) by attending to all positions and taking their weighted average in an embedding space. Non-local Network [35] presented non-local operations as a generic family of building blocks for capturing long-range dependencies. Wang *et al.* [36] proposed to represent videos as space-time region graphs which represents the relationships between humans and objects. Wu *et al.* [37] built a flexible and efficient Actor Relation Graph (ARG) to simultaneously capture the appearance and position relation between actors for group activity recognition. The nonlocal network was also employed in 2s-AGCN [38] for skeleton-based action recognition. However, despite its advantages, the original nonlocal block is infeasible in our method as discussed in Section 3.4. Thus, we design a variant of self-attention: global context adaption module, which directly operates on weights instead of inputs to avoid incompatibility.

Skeleton-Based Action Recognition. There have been a large number of works proposed for action recognition through modeling skeleton dynamics. Early works mainly relies on hand-craft features, e.g., covariance matrices of joint trajectories [39], relative positions of joints [40] or rotations and translations between body parts [41]. Recently, deep learning based models [10], [42], [43], [44] have been prevalent for skeleton modeling, e.g., RNN, CNN and GCN. RNN owns excellent temporal modeling ability, thus have been used in capturing temporal dependency of joints [12], [45], [46]. CNN-based methods [13], [14] transformed the skeletons into a series of color images and fed them into the CNN to classify action category. Temporal CNNs [47], [48], [49] were also proposed to learn skeleton motion. Taking advantages of GCN, recent methods [8], [15], [17], [50] obtained significant improvements, e.g., HCN [10] focused on co-occurrence feature learning framework featuring global information of joints. Despite the progress that previous works achieved, the spatial context of joints is not well exploited.

For local context, some works considered spatial context mostly through hand-craft modules, e.g., body partitions. ST-GCN [8] proposed a general GCN for skeleton modeling, but it also introduced new partition strategies of GCN kernel to handle spatial dependency. Although SR-TSL [9] performed spatial reasoning by decomposing body into several parts to address group-level context, it directly aggregates features by fully-connection (FC) while ignoring inter-group and intra-group relationships. The effects of the skeleton topology (global context) are still neglected.

Recent works have exploited the global-level context by introducing the attention mechanism. AS-GCN [51] combined action and structural links to capture richer dependencies. AGC-LSTM [52] proposed attention-enhanced graph convolutional LSTM to capture high-level semantic features. DGNN [53] represented the skeleton data as a directed acyclic graph (DAG) and designed a novel feature extraction algorithm for DAG. However, although 2s-AGCN [38] has attempted to parameterized the skeleton graph, its backbone is similar to the vanilla GCN-based

network of ST-GCN [8], leading to that 2s-AGCN cannot avoid the limitations of GCN. Most importantly, all above methods lack graph-adaption ability thus cannot handle different skeleton topologies except well-designed one. On the other hand, due to weight-sharing and fixed graph, GCN-based methods cannot capture diverse local context. To address the above problems, we first leverage complement graphs as multi-views and propose unified MV-IGNet to hierarchically capture multi-level spatial skeleton context.

3 MULTI-VIEW INTERACTIONAL GRAPH NETWORK

As motivated in Section 1, we reform the prevalent feature extraction paradigm for skeleton graph and develop a unified interactional graph learning framework for skeleton-based action recognition. Our method named *MV-IGNet*, captures multi-level spatial skeleton context by learning graphs and weights simultaneously. MV-IGNet is the ensemble of two complementary IGNet models. Each IGNet consists of three functional parts: the first is its basic unit, *Separable Parametric Graph Convolution (SPG-Conv)* which enables multiple learnable graphs as parameters to capture richer interactions between joints and their neighbors. Compared to vanilla GCN, SPG-Conv is more efficient and effective, which also has stronger adaption capability to handle irregular graphs. Based on SPG-Conv, *SPG-pooling* divides skeleton joints into several groups to capture the inter-group and inter-group interactions. The global input-dependent graph is also learned by *Global Context Adaption (GCA)* module during inference to adaptively determine the relationships of joints. In MV-IGNet, two IGNet models extract complementary skeleton features by using complement graphs rather than the same graph, which is simple yet effective. We describe each functional part of MV-IGNet and its technical details in the following sections.

3.1 Revisit: GCN for Skeleton Graph

We first construct the spatiotemporal skeleton graph and then revisit several graph convolution networks (GCNs) that are related to our method.

3.1.1 Skeleton Graph Construction

A skeleton sequence with V joints and T frames can be represented as a structured spatiotemporal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. $\mathcal{V}$ is a node-set including all joints in sequence, and $\mathcal{E}$ denotes skeleton edge set including all connections between joints. We adopt the prevalent skeleton graph as described in previous works (e.g., ST-GCN [8]). Specifically, joints are with *physical* connections of the human body in each frame where each joint is connected to the same joint in the consecutive frames. For t th frame where $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$, the neighbor set of the i th joint is defined as $\mathcal{N}_{t,i} = \{v_{t,j} | d(v_{t,i}, v_{t,j}) \leq D\}$, where $d(v_{t,i}, v_{t,j})$ denotes the path distance between the i th and j th joints in graph $\mathcal{G}_t$. We set distance threshold $D = 1$. The topology of graph $\mathcal{G}_t$ can be represented by adjacency matrix $A \in \mathbb{R}^{V \times V}$ where $A_{i,j} \in \{0, 1\}$. In this paper, we mainly focus on learning spatial interactional skeleton graph, i.e., the relationships of joints within the frame. Thus, subscript “ t ” is omitted to simplify formulations, e.g., v_i denotes $v_{t,i}$ without specific explanation.

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3.1.2 Graph Convolution Network

Graph convolution network with fast localized spectral filtering is formulated as follows in [24]:

$$g_\theta(\Lambda) = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\Lambda}), \quad (1)$$

$$y_{c'} = \sum_{c=1}^C g_{\theta_{c,c'}}(L) x_c \in \mathbb{R}^V, \quad (2)$$

where $y_{c'}$ is the c' th output feature map of the input sample, $x_c \in \mathbb{R}^V$ are the input feature maps, C, C' are the number of input and output channels, and V is the number of nodes in graph. $g_\theta(\cdot)$ is the polynomial filter where the parameter $\theta \in \mathbb{R}^K$ is a vector of Chebyshev coefficients and $T_k(\tilde{\Lambda}) \in \mathbb{R}^{V \times V}$ is the Chebyshev polynomial of order k . Thus, in Eq. (1), “spectral filters represented by K th order polynomials of the Laplacian are exactly K -localized” [24].

A localized first-order approximation of above spectral graph convolutions is proposed in [23]:

$$\mathbf{Y} = \tilde{\mathbf{A}} \mathbf{X} \mathbf{W}^{gcn}, \quad (3)$$

where $\mathbf{X} \in \mathbb{R}^{V \times C}$ and $\mathbf{Y} \in \mathbb{R}^{V \times C'}$ are the input and output feature maps respectively. $\mathbf{W}^{gcn} \in \mathbb{R}^{C \times C'}$ is the shared GCN weight for all joints. $\tilde{\mathbf{A}} \in \mathbb{R}^{V \times V}$ is the normalized adjacency matrix, which means Eq. (3) is 1-localized, i.e., path distance $D = 1$.

ST-GCN [8] designs a skeleton-specific graph convolution based on Eq. (3):

$$\mathbf{Y} = \sum_{s=1}^S (\tilde{\mathbf{A}}_s \odot \mathbf{M}_s) \mathbf{X} \mathbf{W}_s^{stgcn}, \quad (4)$$

where different from the K in Eq. (1), S in Eq. (4) is the number of subsets generated by skeleton partition strategy, $\mathbf{M}_s$ is corresponding s th trainable mask to learn the edge importance and $\odot$ is the element-wise multiplication. Compared to Eq. (3), Eq. (4) has S predefined subsets, S trainable masks and corresponding S weights $\{\mathbf{W}_1^{stgcn}, \mathbf{W}_2^{stgcn}, \dots, \mathbf{W}_S^{stgcn}\}$, e.g., $S = 3$ for *spatial configuration partitioning* in ST-GCN.

However, fixed interaction pattern and inflexible shared weights in above GCNs inevitably limit its expressiveness on skeleton modeling. Specifically, *weight sharing* induces that there is no sufficient number of parameters to capture complex/abundant relationships of joints. Eq. (3) uses only $\tilde{\mathbf{A}}$ to represent the joints’ interactions and Eq. (4) also has limited number of parameters $\tilde{\mathbf{A}}_s \odot \mathbf{M}_s$ (i.e., $S = 3$ in ST-GCN) to extract spatial skeleton context in s th subset. The above operators only consider a very small number of interaction patterns. *Fixed graph* in GCN results in that the same interaction pattern is applied to handle different samples of actions, e.g., the graph topologies in Eqs. (3) and (4) are predefined. But, intuitively we can use more suitable action-specific (dynamic) graphs to obtain more representative features for different actions. For instance, directly connecting two hands will make it easier to recognize the action “clap hands” than using *fixed physical topology*. To efficiently

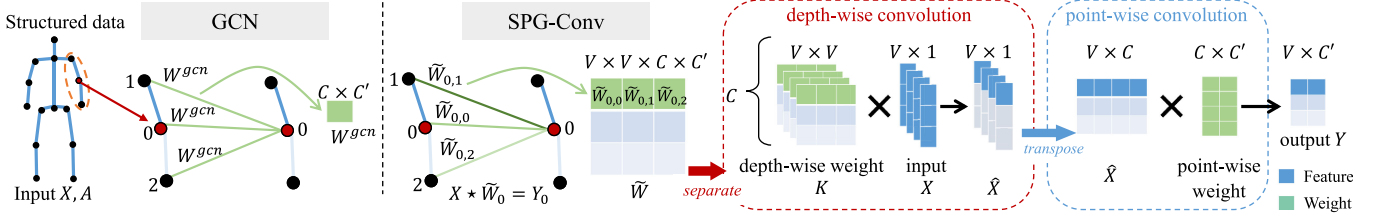


Fig. 2. Separable parametric graph convolution and GCN for the local context modeling. The major difference between SPG-Conv and GCN is whether weight is shared when modeling different interactions. Taking 3-node local graph ($V = 3$) as an example, GCN applies shared weight for all edges while PG-Conv assigns private weights for each edge (e.g., $\tilde{W}_{0,2}$ and $\tilde{W}_{1,3}$). Thus, there are total $3 \times 3 = 9$ private weights with size $C \times C'$. SPG-Conv separates above weight $\mathbf{W} \in \mathbb{R}^{V \times V \times C \times C'}$ into depth-wise graph convolution weight $\mathbf{K} \in \mathbb{R}^{C \times V \times V}$ and point-wise weight with size $C \times C'$, which greatly reduces the memory and computational cost.

address the above problems, we propose the Multi-View Interactional Graph Network (MV-IGNet).

3.2 Learning Local Interactional Graph

In this section, we illustrate how SPG-Conv can efficiently capture joint-level context which is the cornerstone to extract skeleton features. Since each joint possesses specific physical/semantic meaning (different human articulations, e.g., *hand* or *head*), the relationships of joints are usually complex and cannot be comprehensively represented by given adjacent matrix A . Due to that the number of joints V is fixed in each frame (spatial domain), we can readily utilize *private* weight rather than *shared* weight to cope with different interactions of joints, which gives rise to our Separable Parametric Graph Convolution (SPG-Conv).

As illustrated in Fig. 2, given skeleton feature $\mathbf{X} \in \mathbb{R}^{V \times C}$ and joint relationship $A \in \mathbb{R}^{V \times V}$, we replace *shared* weight $\mathbf{W}^{gcN} \in \mathbb{R}^{C \times C'}$ in Eq. (3) by the *private* weight $\mathbf{W}_{i,j} \in \mathbb{R}^{C \times C'}$ to capture the interaction between i th and j th joints (local context)

$$\begin{aligned} \mathbf{Y}_i &= \sum_{v_j \in \mathcal{N}_i} \frac{1}{z_{i,j}} \mathbf{x}_j \mathbf{W}_{i,j} = \sum_{j=1}^V \tilde{\mathbf{A}}_{i,j} \mathbf{x}_j \mathbf{W}_{i,j} \\ &= \sum_{j=1}^V \mathbf{x}_j [\tilde{\mathbf{A}}_{i,j} \mathbf{W}_{i,j}] \\ &= \sum_{j=1}^V \mathbf{x}_j \tilde{\mathbf{W}}_{i,j} \\ &= \mathbf{X}^* \tilde{\mathbf{W}}_i, \end{aligned} \quad (5)$$

where $\tilde{\mathbf{W}}_i \in \mathbb{R}^{1 \times V \times C \times C'}$ captures the local context of i th joint. The final output feature $\mathbf{Y}$ can be computed by

$$\mathbf{Y} = \mathbf{X}^* \tilde{\mathbf{W}} = \mathbf{X}^* [\tilde{\mathbf{A}} \odot \mathbf{W}], \quad (6)$$

where $\odot$ is element-wise multiplication on $[V \times V]$ domain of $\tilde{\mathbf{W}} \in \mathbb{R}^{V \times V \times C \times C'}$, i.e., $\tilde{\mathbf{W}}_{i,j} = \mathbf{0}$ if $\tilde{\mathbf{A}}_{i,j} = 0$. Operation “ $*$ ” can be regarded as parametric graph convolution (PG-Conv) with C input channels and C' output channels, thus the size of each graph convolution kernel is $V \times V \times C$. To further enhance the capability of context modeling, we introduce the practical and efficient variant of PG-Conv, i.e., separable parametric graph convolution (SPG-Conv).

Compared to Eqs. (1) and (2), PG-Conv’s graph filters operated on inputs have similar size $V \times V \times C \times C'$. However, they have totally different number of trainable parameters that are used to capture different interactions in graph.

Eq. (1) is a spectral GCN focusing on K localities while PG-Conv is 1-localized operator on spatial domain. Although the output size of $g_{\theta,c,c'}(L)$ is $V \times V$, Eq. (1) has only K trainable parameters. Thus, Eq. (2) only has $K \times C \times C'$ trainable weights, which are too less to capture abundant relationships of joints in skeleton (K cannot be a large value due to the small size of skeleton graph). Our PG-Conv uses more $V \times V \times C \times C'$ weights to capture abundant 1-localized relationships of joints for discriminating complex action dynamics.

SPG-Conv. In each kernel of PG-Conv, the features of all C channels are directly aggregated (averaged) without distinguishing channel-wise interaction patterns. Considering that skeleton features in different channels (e.g., x, y, z coordinates) may have distinct relationships, we propose SPG-Conv to exploit this information.

As shown in Fig. 2, SPG-Conv shares a similar high-level spirit with separable convolution manner of CNN (e.g., MobileNet [54], Xception [55]). It splits the PG-Conv layer into two parts, a depthwise graph convolution layer for channel-wise filtering and a 1×1 pointwise convolution layer for feature aggregation. For each input channel C , we apply one graph convolution kernel with size $V \times V$ in depthwise graph convolution

$$\hat{\mathbf{X}}_{:,c} = (\tilde{\mathbf{A}} \odot \mathbf{K}_c) \mathbf{X}_{:,c} \quad c = 1, 2, 3, \dots, C, \quad (7)$$

where $\mathbf{X}_{:,c}, \hat{\mathbf{X}}_{:,c} \in \mathbb{R}^{V \times 1}$ are the c th channel of input and output features respectively. $\mathbf{K}_c \in \mathbb{R}^{V \times V}$ is c th graph kernel and $\odot$ is element-wise multiplication. Pointwise convolution, a simple 1×1 convolution, is then used to expand the output of depthwise graph convolution into the other dimension C' . In this case, SPG-Conv has the same input and output size as PG-Conv but captures the channel-wise interaction patterns. It not only drastically reduces computational and memory cost but also further enhance the graph-adaption capabilities of the neural network.

According to Eqs. (3) and (5), PG-Conv owns much more weights than that of GCN as shown in Table 1. The weights of PG-Conv are derived from $\tilde{\mathbf{A}}$ which is usually sparse,

TABLE 1
The Comparison Between GCN and SPG-Conv

Methods	computational cost	#params
GCN	$N_E \times C + V \times C \times C'$	$C \times C'$
PG-Conv	$N_E \times C \times C'$	$N_E \times C \times C'$
SPG-Conv	$N_E \times C + V \times C \times C'$	$N_E \times C + C \times C'$

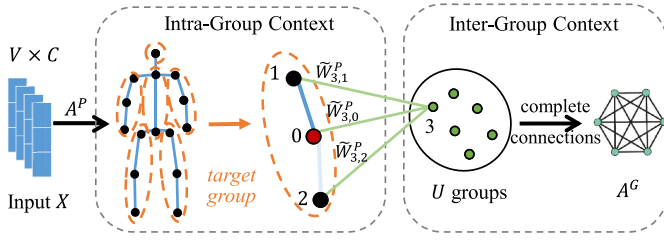


Fig. 3. Hierarchical modeling of interaction context. The skeleton is manually divided into six parts, which can be represented by pooling matrix A^P . For the intra-group context, we first embed V joints into U groups by SPG-pooling. After that, a complete graph A^G is applied to let SPG-Conv learn inter-group context during training.

thus the actual computational cost should be $N_E \times C \times C'$ where N_E is the number of nonzero values in $\tilde{A}$. SPG-Conv splits PG-Conv weights $N_E \times C \times C'$ into $N_E \times C + C \times C'$ which is the sum of the depthwise and pointwise layers, while the corresponding computational cost is $N_E \times C + V \times C \times C'$. We will discuss the efficiency of SPG-Conv in Section 4.4.2.

SPG-Conv enables channel-wise learnable graphs, which can adapt to whole training dataset in training process by back-propagation. This superiority facilitates richer local context and lets SPG-Conv be more compatible with irregular adjacent matrix A . On the other hand, the fixed graph A still limits the flexibility of SPG-Conv, e.g., it is not possible to change the relationships of joints to handle higher-level context.

3.3 Hierarchical Modeling of Interaction Context

Higher-level semantic features [56] are definitely essential to discriminate action representations. However, intra-group (joint relationships within a group) and inter-group (groups' relationships) contexts are not well addressed in previous works. Thus, we take advantages of SPG-Conv to extract richer high-level interaction patterns.

Intra-Group Context Modeling. This step describes how to divide skeleton joints into several groups. SR-TSL [9] has attempted to manually partition the skeleton into five parts. As shown in Fig. 3, we similarly partition body into six parts, i.e., “head, trunk, left arm, right arm, left leg and right leg”. Different from SR-TSL that uses FC layers to encode joints' features within a group, we learn intra-group context by separable parametric graph pooling (SPG-pooling).

Given skeleton feature $X \in \mathbb{R}^{V \times C}$, SPG-pooling aims to pool these V joints into U groups via graph convolution. We use a grouping/pooling matrix $A^P \in \mathbb{R}^{U \times V}$ to represent how

we manually group V skeleton joints into U groups. The element $A_{u,v}^P$ indicates the possibility that v th joint belongs to u th group. In our settings, one joint can only belong to one group, i.e., $A_{u,v}^P \in \{0, 1\}$ and if $A_{u,v}^P = 1$ then $\forall w \neq u, A_{w,v}^P = 0$. To extract intra-group features, we adopt the same solution as that of dealing with the local context, i.e., we apply “private” learnable graph pooling weight $P_{u,v}$ to learn the relationships between v th joint and u th group. Similar to SPG-Conv, SPG-pooling also consists of a depth-wise pooling layer and point-wise convolution layer. The depth-wise pooling applies one graph pooling kernel per channel

$$\hat{X}_{:,c} = (\tilde{A}^P \odot P_c) X_{:,c} \quad c = 1, 2, 3, \dots, C, \quad (8)$$

where $X_{:,c} \in \mathbb{R}^{V \times 1}$, $\hat{X}_{:,c} \in \mathbb{R}^{U \times 1}$ are the c th channel of input and output features respectively. $\tilde{A}^P$ is the normalized pooling matrix A^P , $P_c \in \mathbb{R}^{U \times V}$ is c th graph pooling kernel and $\odot$ is element-wise multiplication. The point-wise convolution layer is the same as that in SPG-Conv. As shown in Fig. 4, SPG-pooling can be regarded as a variant of SPG-Conv with depthwise weight $P \in \mathbb{R}^{U \times V \times C}$ and given graph A^P . Thus, the total parameter number of SPG-pooling is $U \times V \times C + C \times C'$.

Inter-Group Context Modeling. Although SPG-pooling obtains U groups, the interactions among these groups are not determined. To capture the inter-group context, different from defining adjacent matrix to describe groups' interactions, we don't provide any prior information about the relationships of groups, but directly input groups' features into SPG-Conv. The graph configuration $A^G \in \mathbb{R}^{U \times U}$ is set as complete graph (adjacent matrix $A^G = \mathbf{1}$), and with the help of its strong graph-adaption ability, SPG-Conv can learn inter-group context itself in training process. Fig. 4 compares our solutions in dealing with joint-level, intra-group, and inter-group context. We extract the multi-level skeleton context by SPG-Conv in a unified way, which ensures the simplicity and flexibility of neural networks.

3.4 Global Interactional Graph Adaption

As described previously, the fixed graph on spatial domain restricts the interaction path, which makes it insufficient and inflexible to handle complex actions, e.g., adding new edges or selecting input-related edges is infeasible.

Many works [8], [53] have attempted to address above problem by introducing attention. But most of existing attention-based techniques cannot be directly applied in our SPG-Conv, e.g., original self-attention [34] or nonlocal

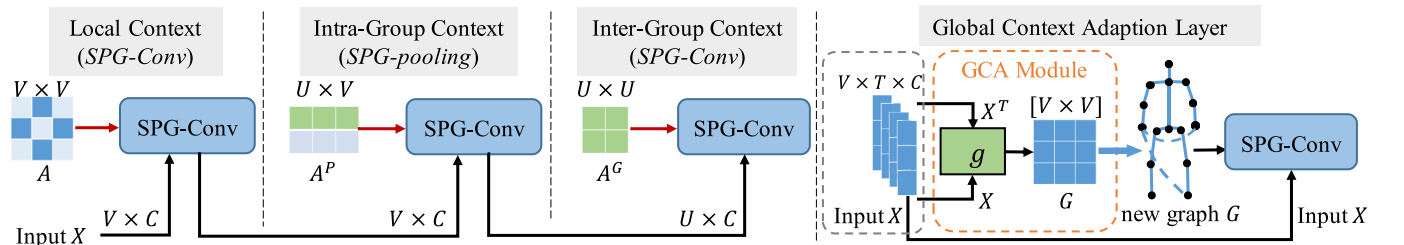


Fig. 4. Proposed unified layers for multi-level skeleton context modeling. All layers receive the output of the previous layer as input. Local, intra-group and inter-group context can be captured by SPG-Conv with different weights and graph settings, i.e., A , A^P and A^G respectively. Graph setting of GCA layer is dynamically generated from input X by GCA module: given input skeleton sequence X , GCA generates input-dependent graph G by Eqs. (9) and (10). We replace the given adjacent matrix $\tilde{A}$ with the learned graph G in SPG-Conv, and perform the element-wise multiplication $G \odot K$ on $[V \times V]$ domain. In this case, the depth-wise weight is updated according to input-dependent graph G in training process.

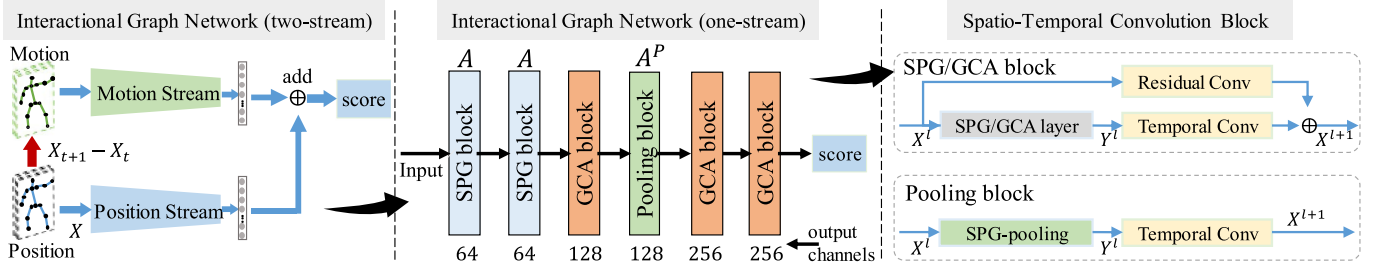


Fig. 5. The two-stream architecture of IGNet. Motion and position streams stack the same 6 spatiotemporal convolution (STC) blocks. STC block is similar to that of ST-GCN but employs our SPG/GCA/SPG-pooling layers rather than GCN to extract spatial features. We don't add residual conv in pooling block due to dimension mismatch.

network [35]. In vanilla CNN, it's no matter for the non-local network to aggregate features before convolutional operations, because CNN's kernels are shared for all positions in the image. However, SPG-Conv kernels are private for each edge, i.e., weight $\tilde{\mathbf{W}}_{i,j}$ is only used to learn the interaction between i th and j th joints. In this case, feature aggregation before SPG-Conv will bring unexpected "noise" for weight $\tilde{\mathbf{W}}_{i,j}$ and further confuse the optimization process. To solve the above problem, we propose global context adaption (GCA) layer which aims to "select" specific input-related weight $\tilde{\mathbf{W}}_{i,j}$ rather than directly weighting features.

Global Context Adaption Module. The objective of GCA is to learn an input-dependent interactional graph. Different from nonlocal/self-attention, given input $\mathbf{X} \in \mathbb{R}^{C \times T \times V}$, we utilize *private* weight $\mathbf{W}_i^\theta, \mathbf{W}_j^\phi \in \mathbb{R}^{C' \times C}$ ($C' = C/2$) to generate the relationship between i th and j th joints

$$g(\mathbf{X}_i, \mathbf{X}_j) = \theta_i(\mathbf{X}_i)^T \phi_j(\mathbf{X}_j) = \sigma(\mathbf{W}_i^\theta \mathbf{X}_i)^T \sigma(\mathbf{W}_j^\phi \mathbf{X}_j), \quad (9)$$

$$G_{i,j} = \frac{1}{\mathcal{C}(\mathbf{X})} g(\mathbf{X}_i, \mathbf{X}_j), \quad (10)$$

where $\mathbf{X}_i \in \mathbb{R}^{C \times T \times 1}$ is the spatiotemporal feature of i th joint and σ reshapes $C' \times T \times 1$ of $\mathbf{W}_i^\theta \mathbf{X}_i$ into $C' \times T \times 1$. The input-dependent graph $G \in \mathbb{R}^{V \times V}$ is normalized by factor $\mathcal{C}(\mathbf{X})$, where $\mathcal{C}(\mathbf{X}) = \sum_{j=1}^V g(\mathbf{X}_i, \mathbf{X}_j)$ and V is the number of joints. As discussed above, graph G cannot be applied on feature $\mathbf{X}$ directly, i.e., $G\mathbf{X}$ in GCN. In the following, we will show how to apply G on SPG-Conv weights $\mathbf{W}$ to construct GCA layer.

Global Context Adaption Layer. There is a crucial difference between GCA and non-local in constructing the convolution block. As shown in Fig. 4, the ultimate goal of GCA is to adaptively "select" specific weight $\tilde{\mathbf{W}}_{i,j}$ in SPG-Conv rather than directly weighting features. For the input spatiotemporal feature $\mathbf{X} \in \mathbb{R}^{C \times V \times T}$, GCA layer first obtains input-dependent graph G by Eqs. (9) and (10). After that, this learned graph G is used as a new skeleton topology (weighted graph) to dynamically replace the graph setting $\tilde{\mathbf{A}}$ of SPG-Conv in Eq. (5)

$$\mathbf{Y}_i = \sum_{j=1}^V G_{i,j} \mathbf{X}_j \mathbf{W}_{i,j}. \quad (11)$$

In GCA layer, SPG-Conv weight $\mathbf{W}_{i,j}$ is updated only when it's selected by $G_{i,j}$ in training process. Thus the selective manner of GCA layer jointly optimizes weight selection (global context) and weight values. In the implementation,

we first perform element-wise multiplication between G and SPG-Conv's depthwise weight $\mathbf{K}$ and then feed input feature $\mathbf{X}$ into updated SPG-Conv to get the final output.

Fig. 4 illustrates the relationships between GCA and proposed other layers. We implement these layers in a unified manner which is very simple yet efficient. They both take SPG-Conv as a basic unit but receive different graphs as input to learn the spatial context of different levels. Thus, it's very easy to construct an efficient framework, interactional graph network (IGNet), for skeleton-based action recognition as shown in Section 3.5.

3.5 Interactional Graph Network

As shown in Fig. 5, IGNet is a two-stream network. Each stream consists of several stacked spatiotemporal convolution (STC) blocks which are used to hierarchically capture spatiotemporal concurrence skeleton features.

Spatio-Temporal Convolution (STC) Block. The proposed STC block is similar to ST-GCN [8] except that we replace the GCN by our SPG-Conv/SPG-pooling/GCA layers on the spatial domain. Given input skeleton feature $\mathbf{X} \in \mathbb{R}^{C \times T \times V}$, we first perform spatial convolution. After that, the temporal context is captured by a vanilla 2D convolution with kernel size $t \times 1$. We also employ temporal separable convolution as described in MobileNet [54] to ensure the balance of parameter number on spatial and temporal domains. A residual connection is added in each block to stabilize the training process when the block is SPG-Conv or GCA block. For SPG-pooling block, we omit residual connection to avoid the dimension mismatch.

Two-Stream. We adopt the two-stream architecture of which the effectiveness has been proved in video-based action recognition tasks. The position stream considers the location information while the motion stream is designed to model the action dynamic cues. These two features are complementary to each other in action recognition. Position stream inputs the coordinates of skeleton sequence $\mathbf{X}$, while the input of motion stream is the displacement of skeleton joints, which can be calculated by two consecutive frames, i.e., $\mathbf{X}_{t+1} - \mathbf{X}_t$. We add a data BatchNorm layer for each stream to normalize input data and perform global spatiotemporal pooling after the last layer to obtain final features. The output scores of two streams are averaged after *softmax* layer to predict actions.

Architecture Settings. Each stream stacks 6 STC blocks. It is very shadow when compared with previous GCN-based models (e.g., 9 blocks in ST-GCN [8]). Output channels of blocks are set as 64, 64, 128, 128, 256 and 256 respectively.

The first 2 blocks are SPG-Conv layers with graph setting A , which are used to encode sufficient features for GCA. Then, the 3rd GCA layer can select proper weights which discovers latent significant interactions beyond the given A . The followed SPG-pooling block reduces the number of joints from V to U (usually $U < V$). After that, two GCA block are used to learn the relationship of groups, in which SPG-Conv receives $U \times U$ complete graph. In this way, the multi-level spatial context is captured by IGNet.

3.6 From Single View to Multiple Views

Although IGNet can automatically learn the global topologies from inputs, it faces a dilemma that the search space of GCA is too large to find the optimal topology. GCA needs to estimate the importance of all possible $V \times V$ candidates/edges, but features usually cannot guarantee this large space searching. Too many “noisy” edges for each joint will confuse the optimization process (See Section 4.4.4). To alleviate the above weaknesses, we further enhance the ability to learn joint dependency by generating multi-view features.

Generally, different camera views can provide complementary information of skeleton, which can be represented as affine transformations on original coordinates, i.e., C dimension of skeleton data $\mathbf{X} \in \mathbb{R}^{C \times V}$. We can define different graphs as transformations on V dimension of $\mathbf{X}$, i.e., graph convolution. Similar to the “isomers” in chemistry, two skeleton graphs that have the same joints but different topologies can reflect different properties of the human skeleton. Thus, graph topology is also a critical perspective to offer complementary information. When we use different graphs to cooperatively represent the same action, features generated from different topologies should be complementary. In this way, we propose the multi-view interactional graph network (MV-IGNet) to extract features with different graphs.

Multi-View Generation. We first show how to get different graphs for multi-view learning. Given spatial skeleton topology A^0 (physical), we expect to use a different topology A^1 to construct a new skeleton structure for the same data $\mathbf{X}$. Intuitively, there are two fundamental requirements for A^1 : 1) *Complementarity*: the A^1 should possess as much complementary information as possible, which indicates lower similarity between A^0 and A^1 . 2) *Feasibility*: the A^1 that doesn’t harm the performance of our method, should be easily found. Therefore, due to the strong graph-adaption capability of SPG-Conv, we adopt the simplest approach to obtain different topologies by finding the complement/inverse graph of A^0 . The complement A^1 of a graph A^0 is a graph on the same joints such that two distinct joints of A^1 are adjacent if and only if they are not adjacent in A^0 . Namely, these two graphs have no common edges. In implementation, we get A^1 by

$$A^1 = \mathbf{1} - A^0, \quad (12)$$

where A^1 is the adjacent matrix of complement graph of A^0 , and $\forall A^0_{i,j} \in \{0, 1\}$, $A^0_{i,j} + A^1_{i,j} = 1$. A^1 is a complementary way to describe the skeleton and SPG-Conv can discover optimal interaction patterns within A^1 in training process.

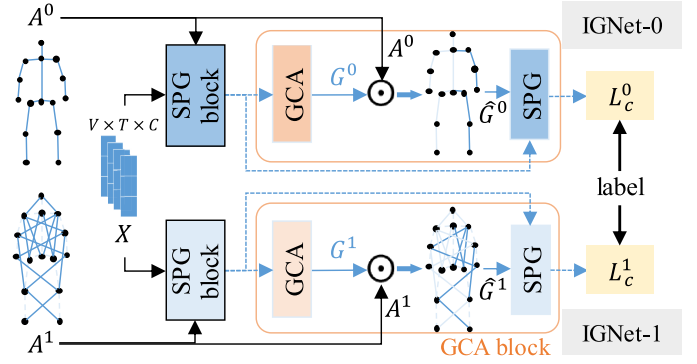


Fig. 6. MV-IGNet. It consists of two ensemble IGNet models with different graphs (A^0 and A^1). We only show some necessary layers in MV-IGNet. These two models are trained independently and their prediction scores are averaged to get final results for the test.

Multi-View Interactional Graph Network. MV-IGNet targets at using above generated multi-views, A^1 and A^0 , to extract complementary features. As shown in Fig. 6, MV-IGNet is an ensemble model consisting of two IGNet models which are trained independently except being given different graphs (A^0 and A^1 respectively). Each model takes the same position and motion data as inputs, and receive the same label to supervise the action classification. For the test, we simply average the prediction scores of two IGNet models. Different from the GCA in IGNet, search spaces of GCA modules in MV-IGNet are constrained by two graphs A^1 and A^0 respectively

$$\hat{G}^m = A^n \odot G^m, \quad (13)$$

$$\mathbf{Y}_i = \sum_{j=1}^V \hat{G}_{i,j}^m \mathbf{X}_j \mathbf{W}_{i,j}, \quad (14)$$

where G is the input-dependent graph generated by Eq. (10), $\hat{G}$ is the constrained graph (see Figs. 4 and 6), $\odot$ is element-wise multiplication and n denotes the n th IGNet model in MV-IGNet. In this case, the search spaces are conditioned on the given graph A^n to avoid topology conflicts and ensure complementary information. Thus, parameter numbers of two models are slightly different because valid number $N_{\mathcal{E}}^1$ of graph A^1 is larger than that of A^0 .

Note that SPG-Conv has strong adaption ability to handle many irregular topologies that other methods don’t have (see Section 4.4.2). This superiority guarantees the high performance of IGNet with large graph A^1 . Therefore, MV-IGNet possesses strong robustness and can dramatically improve the action recognition performance.

4 EXPERIMENTS

4.1 Datasets and Implementation

NTU-RGB+D. This dataset [12] contains total 56,000 video clips with 60 action classes in total, which are divided into three major groups: 40 daily actions, 9 health-related actions, and 11 mutual actions. We adopt two standard evaluation protocols defined in [12] for this dataset: Cross-Subject (CS) and Cross-View (CV). In Cross-Subject protocol, the 40 subjects are split into training and testing groups. Each group consists of 20 subjects. For Cross-View

TABLE 2
Accuracy and Efficiency Comparisons on Cross-Subject (CS) and Cross-View (CV) Protocols of NTU-RGB+D Dataset

No.	Method	Separable	SPG	Pool	GCA	MV	CS	CV	#time (ms)	#params (M)
0	ST-GCN [8]						81.5	88.3	3.426	3.103
	2s-AGCN [38]						88.5	95.1	8.862	6.940
1	ST-GCN*						81.7	91.2	6.851	6.206
	S-ST-GCN*	✓					80.9	89.8	4.328	2.048
2	SPGNet	✓	✓				86.8	94.3	0.845	0.848
3	HPGNet	✓	✓	✓			87.2	94.7	0.553	0.793
4	IGNet	✓	✓	✓	✓		87.5	94.9	0.815	0.887
5	MV-HPGNet	✓	✓	✓		✓	88.6	95.8	1.106	1.650
6	MV-IGNet	✓	✓	✓	✓	✓	88.8	96.1	1.630	1.838

We show the architecture difference of our baselines and also report their inference time (ms) and parameter number (M) to demonstrate the advantages of our method. MV-IGNet can achieve satisfactory results with smaller model size and faster inference.

evaluation, all the samples of the camera 2 and 3 are used for training and samples of camera 1 are used for testing.

NTU-RGB+D 120. NTU-RGB+D 120 dataset [57] extends above NTU RGB+D dataset [12] by adding another 60 classes and another 57,600 video samples, i.e., NTU RGB+D 120 has 120 classes and 114,480 samples in total, which are divided into three major groups, including 82 daily actions, 12 health-related actions, and 26 mutual actions. We adopt two standard evaluation protocols defined in [57] for this dataset: Cross-Subject (CSub) and cross-setup (CSet). For cross-subject evaluation, the 106 subjects are split into training and testing groups. Each group consists of 53 subjects. For cross-setup evaluation, we pick all the samples with even collection setup IDs for training, and those with odd setup IDs for testing, i.e., 16 setups are used for training, and the other 16 setups are reserved for testing.

Northwestern-UCLA. This dataset [58] contains 1,494 video clips covering 10 categories and is captured by three Kinect cameras simultaneously from a variety of view-points. Each action sample contains RGB+D and human skeleton data performed by 10 different subjects. The evaluation protocol is the same as [58]. Samples from the first two cameras constitute the training set and samples from the other camera constitute the testing dataset.

Implementation Details. We change the width of the neural network by width multiplier α . Without specific explanation, the settings of IGNet are shown in Fig. 5, i.e., width multiplier $\alpha = 1.0$ and skeleton graph configuration is *physical A⁰*. In experiments, we resize skeleton sequence into a fixed-length T as input on temporal dimension, and set $T = 200, 200$ and 75 for NTU, NTU-120, and UCLA respectively. Motion data is generated by calculating the displacement of coordinates between consecutive 1 frames, i.e., $\mathbf{X}_{t+1} - \mathbf{X}_t$. Global pooling is applied to obtain final features followed by a fully-connected layer for classification. During training, IGNet is optimized by SGD for 120 epochs with initial learning rate 0.2 decayed by 0.1 after [40, 80, 100] epoch. Dropout with a probability of 0.5 is adopted to avoid over-fitting. Batch sizes for the NTU&NTU-120 and UCLA are 64 and 24 respectively. Width multiplier α is set as 1.0 for NTU&NTU-120 and 0.5 for UCLA dataset.

4.2 Baselines

We introduce following baselines for comparison and their differences are shown in Table 2.

1. **ST-GCN***: for a fair comparison, similar to IGNet, we build two-stream ST-GCN [8] by adding motion branch. ST-GCN is the original method in [8], ST-GCN* is its two-stream variant, and S-ST-GCN* is the variant of ST-GCN* with separable convolution.
2. **SPGNet**: it adopts 6-block two-stream architecture which replaces SPG-pooling and GCA blocks in IGNet with SPG-Conv blocks.
3. **HPGNet**: it replaces GCA blocks in IGNet with SPG-Conv blocks. The number of joints is reduced from V to U in HPGNet.
4. **IGNet**: this is proposed interactional graph network (IGNet) as shown in Fig. 5.
5. **MV-HPGNet**: this is the multi-view variant of HPGNet. We build it by giving two complementary graphs to ensemble HPGNet models.
6. **MV-IGNet**: this is the complete version of MV-IGNet with ensemble IGNet models as shown in Fig. 6.

4.3 Ablation Study

Extensive ablation studies are performed to validate the effectiveness of each contribution. Table 2 reports the results of all baselines on NTU-RGB+D dataset (cross-subject and cross-view protocols).

Two-Stream. Table 3 compares the performance of SPGNet models with motion and position streams respectively. The two-stream model outperforms one-stream-based models.

SPG-Conv. We compare the results of ST-GCN*, S-ST-GCN*, and SPGNet on NTU-RGB+D dataset in Table 2. We set width multiplier $\alpha = 1.0$, i.e., the output channels are set according to Fig. 5. Fig. 7 evaluates the effects of network width, which shows the results of above baselines with different channel width, i.e., $\alpha = 0.5, 1.0, 1.5$, and 2.0 respectively. Note that joints are connected according to *physical*

TABLE 3
Accuracy (%) of SPGNet Models on NTU-RGB+D

method	configuration	CS	CV	#params
SPGNet	<i>two stream</i>	86.8	94.3	0.848M
SPGNet	<i>position</i>	84.1	92.4	0.424M
SPGNet	<i>motion</i>	83.8	91.8	0.424M

$\alpha = 1.0$

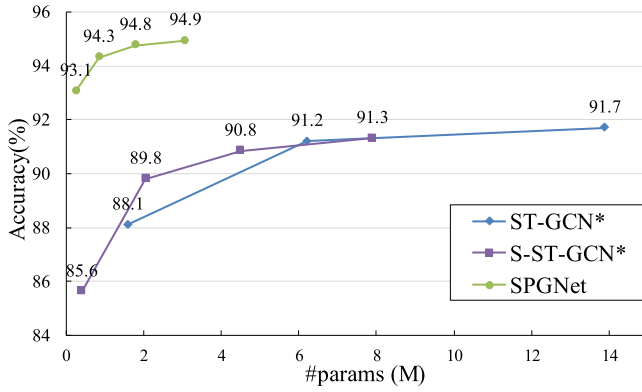


Fig. 7. #Param-accuracy curve on NTU-RGB+D (CV). For S-ST-GCN* and SPG-Conv, width multiplier $\alpha = 0.5, 1.0, 1.5, 2.0$. The basic output channels of ST-GCN are $\{64, 64, 64, 128, 128, 128, 256, 256, 256\}$ ($\alpha = 1.0$) and we only plot $\alpha = 0.5, 1.0, 1.5$ for ST-GCN. The performance of baselines has an obvious positive correlation with the parameter number. SPGNet outperforms ST-GCN on both efficiency and accuracy.

relationships for all baselines in Table 2 and Fig. 7. The number of parameters is calculated according to the number of nonzero values N_E . SPG-Conv uses fewer parameters than ST-GCN* but outperforms ST-GCN* by a large margin and #param-accuracy curve also shows SPG-Conv is more efficient and effective than GCN.

SPG-Pooling. Comparing the results of SPGNet (*w/o* SPG-pooling) and HPGNet (*w/* SPG-pooling) in Table 2, we show that HPGNet uses fewer weights but achieves stably better accuracy than SPGNet on both CS and CV protocols, which proves the significance of higher-level context.

GCA Module. Table 2 compares the results of HPGNet (*w/o* GCA module) and IGNet (*w/* GCA module). IGNet outperforms HPGNet by about 0.3 percent in both CS and CV protocols, which indicates adaptively selecting input-dependent interactions can also help to recognize actions.

Multi-Views. In Table 2, compared to IGNet and HPGNet, MV-IGNet and MV-HPGNet obtain the significant improvements, showing the vital contribution of complementary views.

4.4 Model Analysis

We first show the overall memory and computational efficiency of our method and then perform the robustness analysis of each module respectively.

4.4.1 Memory and Computational Efficiency

The experimental settings are described in Section 4.1. We use the average inference time (*ms*) per sample to represent

the running speed, which is evaluated on single Nvidia RTX 2080Ti and Pytorch platform. Table 2 reports the inference time and parameter number of different methods, proving that our method is indeed efficient. Our HPGNet is the fastest model with the least parameter number, and MV-IGNet can obtain higher accuracy.

Many recent works [38], [51], [53] are based on ST-GCN [8] but introduce additional parameters and computational cost. These additional modules introduce accuracy gain but inevitably limits the model efficiency. On the contrary, our method can reduce memory cost and use less time to infer while achieving remarkable performance. Taking 2s-AGCN as example, its inference speed is $8.862 = 4.431 \times 2 \text{ ms} / \text{sample}$ while that of our MV-IGNet is $1.630 \text{ ms} / \text{sample}$. Our method's computational efficiency majorly comes from two aspects: 1) the low computational cost of SPG-Conv that reduces the inference time of each layer; 2) the shadow hierarchical network that reduces the overall size of feature maps then increases the running speed.

4.4.2 Separable Parametric Graph Convolution

SPG-Conv not only can obtain higher accuracy with fewer weights but also owns strong graph-adaption capability that GCN doesn't have. Namely, it can achieve high performance even though being given irregular skeleton topologies.

SPG-Conv versus GCN. SPG-Conv sets multi-graphs as parameters to enrich local context, which ensures its special superiority. While ST-GCN is a GCN-based model with hand-craft skeleton graph partition algorithm, which splits the neighbor set into 3 subsets, thus employs 3 independent GCN weights to capture spatial features. According to Table 1, compared to ST-GCN, SPG-Conv reduces the number of parameters by

$$\frac{N_E \times C + C \times C'}{3 \times C \times C'} = \frac{N_E}{3C'} + \frac{1}{3}, \quad (15)$$

where for NTU-RGB+D dataset ($V = 25, E = 24$), $N_E = V + 2E = 73$ and $C' \geq 64$, SPG-Conv uses at most 72 percent weights of ST-GCN on spatial domain (52.3 percent for $C' = 128$).

The Effects of Graph Configurations in SPG-Conv Layer. To evaluate it, we introduce following skeleton graph configurations as shown in Fig. 8: 1) *joint*: a graph only has self-loops, without connections between arbitrary two joints; 2) *edge*: a graph ($A^0 - I$) has physical connections but without self-loops; 3) *physical*: a graph (A^0) has physical connections; 4) *complete*: complete graph, i.e., $\forall A_{i,j} = 1$.

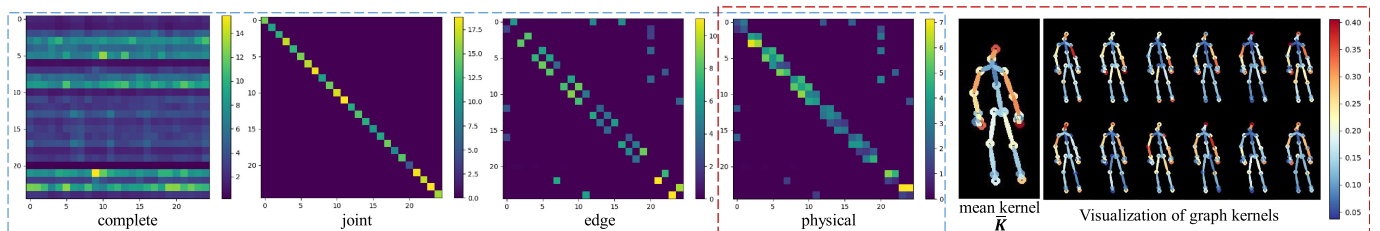


Fig. 8. Visualizing the weights of SPG-Conv. The left shows the different graphs of skeleton: *complete*, *joint*, *edge* and *physical* respectively. Values in these graphs are the summation of graph kernels K in SPG-Conv. The right colorful skeletons show the weights of SPGNet-*physical* in skeleton form. Joints and edges show different importance learned by SPG-Conv. Zoom in for the best visualization.

TABLE 4
Comparison Test on the NTU-RGB+D

Method	configuration	CS	CV
ST-GCN*	<i>joint</i>	78.6	87.1
	<i>edge</i>	81.4	90.8
	<i>physical</i>	81.7	91.2
	<i>complete</i>	73.3	83.2
SPGNet	<i>joint</i>	85.5	93.2
	<i>edge</i>	86.8	94.2
	<i>physical</i>	86.8	94.3
	<i>complete</i>	86.6	94.2
HPGNet	<i>manual</i> A^P	87.1	94.7
	$A^P = \mathbf{1}$	87.2	94.7
IGNet	<i>nonlocal</i>	87.0	94.5
	GCA	87.5	94.9

$\alpha = 1.0$.

TABLE 5
Accuracy (%) on NTU-RGB+D

Stage	-	1	2	3	4	5	6
CV	94.3	94.0	94.3	94.7	94.7	94.6	94.5

SPG-pooling block is insert into different stages of HPGNet models and "-" indicates the SPGNet model without pooling. $\alpha = 1.0$.

Table 4 compares the adaptive capability of ST-GCN* and SPGNet with different topologies of skeleton graphs. For *joint* setting, the decline in accuracy of SPGNet is less than that of ST-GCN*. SPGNet-*edge* achieves nearly the same accuracy with SPGNet-*physical*. Above observations indicate that compared to self-loops, interactions among joints are more important in representing actions. Note that *complete* setting does not contain any prior information. SPGNet-*complete* performs similarly as SPGNet-*physical*, which demonstrates the ability of SPG-Conv to extract complementary features from other graphs (beyond *physical* connections). In summary, all SPGNet models perform satisfactorily and ST-GCN* fails in some extreme cases, which reflects the strong adaption ability of SPG-Conv.

4.4.3 Separable Parametric Graph Pooling

In this section, we mainly explain why we choose hand-craft partitions for groups rather than letting SPG-pooling learn to group partitions itself.

Which Stage to Add SPG-Pooling Layer? Table 5 compares a single SPG-pooling block on the different stages of HPGNet. The improvements of the SPG-pooling on 3, 4, 5, 6th layers are obvious while pooling on the first layer will reduce the overall accuracy. One possible explanation is that features on shadow layers cannot well represent spatiotemporal concurrence features thus are not sufficient to provide precise semantic information. Thus we set 4th block as SPG-pooling block in the following experiments.

The Effects of Group Number U in SPG-Pooling Layer. We show the results of HPGNet when SPG-pooling receives a complete graph $A^P = \mathbf{1}$ instead of hand-craft pooling matrix. Table 6 reports the relationships between accuracy

TABLE 6
Results (%) of HPGNet Models With Different U on NTU-RGB+D

U	25	20	15	10	6	1
CV	94.3	94.4	94.6	94.7	94.7	94.1

TABLE 7
Accuracy (%) on NTU-RGB+D

w/o stages	-	1	{1,2}	{1,2,3}
CV	94.9	94.9	94.9	94.7

GCA is removed in some stages of IGNet.

and U . The joint number of NTU-RGB+D dataset is $V = 25$, thus we set $U = 20, 15, 10$ and 6 respectively to choose a relatively better value of U . The 4th block is set as SPG-pooling block in HPGNet. When $U = V = 25$, HPGNet is same as SPGNet. For $\forall U < V$ except $U = 1$, HPGNet outperforms SPGNet, which means higher-level context is essential in representing actions.

The Effects of Graph Configurations in SPG-Pooling Layer. The hand-craft approach partitions joints into 6 groups as shown in Fig. 3. Table 4 compares SPG-pooling with *manual* or *complete* graphs when $U = 6$. There is no clear difference in performance of HPGNet models in Table 4, but HPGNet ($A^P = \mathbf{1}$) has more parameters than HPGNet-*manual*. Considering both accuracy and computational cost, we adopt manual partitions for SPG-pooling in all the following experiments.

4.4.4 Global Context Adaption Module

Which Stage to Add GCA Layer? In IGNet, we set the first 2 blocks as SPG-Conv rather than GCA blocks for the following reason: Table 7 shows that this modification doesn't harm final performance. The accuracy drops to 94.7 only when we set {1,2,3} stages as SPG-Conv blocks in IGNet.

GCA versus Nonlocal. To prove the advantages of GCA, we replace GCA modules by nonlocal blocks [35] in IGNet for comparison. As discussed in Section 3.4, different from the original dot-product version of nonlocal block proposed in [35], GCA doesn't aggregate features before convolution but adopts weight-selection manner. Table 4 compares the results of IGNet-*nonlocal* and IGNet-GCA. Due to the shared-weight mechanism, our IGNet with nonlocal blocks cannot obtain satisfactory results and IGNet-GCA can improve accuracy to some degree. That's the reason why we devise the novel GCA module for SPG-Conv. Examples are also visualized in Fig. 9 to show how GCA module works (see Section 4.4.5).

4.4.5 Multi-Views

To comprehensively evaluate the proposed MV-IGNet, we introduce following additional baselines in Table 8: 1) *Wide-HPGNet*: this is wider HPGNet (width multiplier $\alpha = 2.0$). 2) *Ensemble HPGNet*: this is ensemble HPGNet (HPGNet | HPGNet in Table 8) in which two HPGNet models have the same graph A^0 . 3) *Single view ST-GCN* | HPGNet* ($A^0 | A^0$):

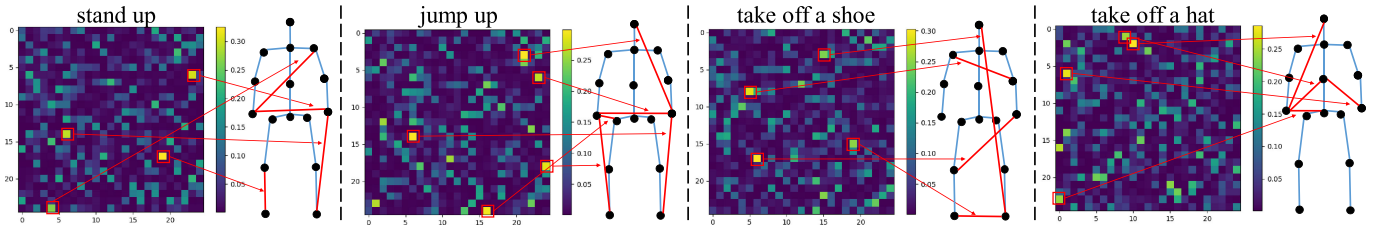


Fig. 9. Visualizing the input-dependent graphs G learned by GCA modules on NTU-RGB+D dataset. Joint number $V = 25$ and its configuration is shown in [57]. For better understanding, we simplify the skeleton graph in Fig. 8 by grouping joints in hand and foot as one joint. Four graphs corresponding to different actions are shown. We use “red” lines to point out interactions with high importance. Zoom in for the best visualization.

this is an ensemble network with same graph A^0 but different backbones. 4) *Multi view ST-GCN* | HPGNet ($A^0 | A^1$)*: this is a multi-view network with different graphs and different backbones.

Multi-Views versus Ensemble. Multi-view baselines outperform all of the other methods by a large margin. Especially, MV-HPGNet performs better than ensemble HPGNet while ensemble HPGNet obtains nearly the same results with Wide-HPGNet. Considering the results of Wide-HPGNet and HPGNet | HPGNet in Table 8, the performance gain of MV-HPGNet mainly comes from multi-views rather than the number of parameters.

The Generality of Multi-View. Most importantly, we combine ST-GCN* and HPGNet models with or without multi-views to validate whether MV is general among graph-based methods. We set physical connection A^0 as the graph of ST-GCN* and give its complement graph A^1 to HPGNet. Table 8 also shows the advantages of MV, which can generally improve the performance across graph-based methods by integrating complementary features.

4.4.6 Qualitative Results

Static Interaction Patterns Learned by SPG-Conv. Fig. 8 shows the weights of SPG-Conv layer in SPGNet with *complete*, *joint*, *edge* and *physical* graphs. We visualize the mean of learned graph kernel $\bar{\mathbf{K}} = \frac{1}{C} \sum_{c=1}^C \mathbf{K}_{:,c}$ in matrix form and also show each channel $\mathbf{K}_{:,c}$ of SPGNet-physical in the form of skeleton. SPGNet-complete learns the relative importance of both joints and edges. SPGNet-physical gives *arms*, *head* and *legs* high importance. Most importantly, it can learn more abundant interaction patterns than the baseline (ST-GCN has only $S = 3$ masks), *i.e.*, it obtains a set of diverse weighted graphs which indicates different interaction patterns. Since each SPG-Conv layer has C graph filters with size $V \times V$ where C is much larger than S in Eq. (4), the combination of more abundant patterns ensures a stronger

feature extraction ability, which indeed guarantees better accuracy and smaller model size than baseline. The weakness of fixed graph is also reported by Fig. 8 that graph-based methods are limited by fixed skeleton topology which restricts the information flow path (interactions between joints). Thus SPG-Conv can only learn the importance within the predefined topology, which is not flexible and powerful.

Dynamic Interaction Patterns Learned by GCA. Fig. 9 shows the input-dependent graphs G learned by GCA module of IGNet in both matrix and skeleton forms. Compared to SPGNet with fixed graph topology in Fig. 8, Fig. 9 illustrates that IGNet can learn more reasonable interactions beyond the predefined one. For instance, for the action “take off a hat”, it learns new interactions between *hand* and *head* that *physical* graph doesn’t have, which can better represent the relationships between head and hands. Moreover, person usually needs to bend down to accomplish the action “take off a shoe”, thus GCA module gives high importance to the *hand*↔*foot* and *head*↔*foot* interactions. Action “jump up” and “stand up” may be similar at the beginning, thus GCA module learns similar interaction patterns for these two actions. On the other hand, Fig. 9 also reveals the weakness of the GCA module that too many edges are selected, which may increase the difficulty of optimization.

4.5 Comparison With the State-of-the-Arts

NTU-RGB+D. According to Fig. 7, SPGNet approaches the upper-bound of performance when width multiplier $\alpha \geq 1.5$. Thus, Table 9 reports the results of MV-IGNet with $\alpha = 1.0$ and 1.5 for comparison. MV-IGNet ($\alpha = 1.5$) outperforms MV-IGNet ($\alpha = 1.0$) by about 0.4 and 0.2 percent accuracy on CS and CV protocols respectively. MV-IGNet ($\alpha = 1.5$) achieves the best accuracy on CV protocol and gets comparable results with the state-of-the-art on CS protocol.

Compared with others, MV-IGNet outperforms current best LSTM based method AGC-LSTM [52] on both CS and CV. ST-GCN [8] is mostly relative to ours. We obtain huge improvements compared to ST-GCN, which shows the importance of spatial interaction context again. Compared to CNN-based method HCN [10], SPGNet surpasses it by about 0.3 and 2.8 percent on CS and CV respectively. AS-GCN [51], AGC-LSTM [52] and 2s-AGCN [38] are GCN-based methods with different attention mechanisms, and their architectures are based on ST-GCN [8] but more complex, which means their parameter numbers and computational costs are larger than that of ST-GCN. Thus our MV-IGNet surpasses them in both accuracy and efficiency. DGNN [53] designs a directed

TABLE 8
Accuracy (%) on NTU-RGB+D

Method	conf.	CS	CV	#params
Wide-HPGNet	$A^0, \alpha = 2.0$	87.3	94.9	2.960M
HPGNet HPGNet	$A^0 A^0$	87.4	95.0	1.586M
ST-GCN* HPGNet	$A^0 A^0$	86.8	94.9	6.999M
ST-GCN* HPGNet	$A^0 A^1$	87.3	95.3	7.031M
MV-HPGNet	$A^0 A^1$	88.6	95.8	1.650M
MV-IGNet	$A^0 A^1$	88.8	96.1	1.838M

“ $A^0 | A^1$ ” means two models trained with graphs A^0 and A^1 independently.

TABLE 9
Action Recognition Accuracy (%) on NTU-RGB+D

Method	CS	CV	Year
HBRNN-L [59]	59.1	64.0	2015
PA-LSTM [12]	62.9	70.3	2016
TG-LSTM-TS [45]	69.2	77.7	2016
Two stream RNN [60]	71.3	79.5	2017
Temporal Conv [49]	74.3	83.1	2017
C-CNN + MTLN [48]	79.6	84.8	2017
VA-LSTM [42]	79.4	87.6	2017
ST-GCN [8]	81.5	88.3	2018
DPRL+GCNN [15]	83.5	89.8	2018
HCN [10]	86.5	91.1	2018
SR-TSL [9]	84.8	92.4	2018
AS-GCN [51]	86.8	94.2	2019
AGC-LSTM [52]	89.2	95.0	2019
2s-AGCN [38]	88.5	95.1	2019
4s-DGNN [53]	89.9	96.1	2019
MV-IGNet ($\alpha = 1.0$)	88.8	96.1	-
MV-IGNet ($\alpha = 1.5$)	89.2	96.3	-

graph neural network (DGNN). The two-step updating of DGNN, i.e., vertex-updating followed by edge-updating process, is a little bit complex and not straightforward, and the input data of DGNN is also different from other mainstream methods. It's difficult to directly compare our MV-IGNet with DGNN. MV-IGNet outperforms DGNN on CV protocol but gets lower accuracy than DGNN on CS protocol. Compared to DGNN, the strength of MV-IGNet lies in the easy implementation, small model size, and fast inference while with comparable accuracy.

NTU-RGB+D 120. This dataset is newly published and most of the existing methods have not been tested on this new dataset yet. For convincing comparison, we not only compare the results reported in NTU RGB+D 120 [57] but also test all aforementioned baselines in Section 4.2. We follow the test protocol of NTU-RGB+D 120 [57] which is different from that of NTU-RGB+D [12], i.e., we adopt *cross-setup* instead of *cross-view* protocol for test as described in Section 4.1. Results in Table 10 clearly show the superiority of the proposed method that MV-IGNet outperforms all baselines and achieves the state-of-the-art accuracy.

Northwestern-UCLA. As shown in Table 11, the proposed MV-HPGNet again achieves the best accuracy of 93.3 percent on the UCLA dataset, surpassing the previous

TABLE 10
Action Recognition Accuracy (%) on NTU-RGB+D 120

Method	C-Subject	C-Setup	Year
Two stream RNN [60]	61.2	63.3	2018
MT-CNN [61]	62.2	61.8	2018
Pose Evolution Map [62]	64.6	66.9	2018
ST-GCN*	75.9	78.2	2018
SPGNet	80.9	82.4	-
HPGNet	81.3	83.1	-
IGNet	81.5	83.4	-
MV-HPGNet	83.5	85.4	-
MV-IGNet	83.9	85.6	-

TABLE 11
Action Recognition Accuracy (%) on UCLA Dataset

Methods	Accuracy (%)	Year
Lie group [41]	74.2	2014
Actionlet ensemble [63]	76.0	2014
HBRNN-L [59]	78.5	2015
Visualization CNN [13]	86.1	2017
Ensemble TS-LSTM [64]	89.2	2017
AGC-LSTM [52]	93.3	2019
ST-GCN*	88.1	2018
SPGNet	91.3	-
HPGNet	91.5	-
IGNet	91.3	-
MV-HPGNet	93.3	-
MV-IGNet	93.1	-

state-of-the-art methods. The performance of baselines on UCLA shows some differences with that on NTU dataset, i.e., IGNet fails to get better results than HPGNet. One possible reason is the lack of training data: about 1,500 video clips of UCLA compared to 56,000 clips of NTU-RGB+D, which cannot guarantee IGNet's learning skeleton topologies and features simultaneously.

5 CONCLUSION

This paper introduces a multi-view interactional graph network (MV-IGNet) for skeleton-based action recognition. The proposed IGNet can adaptively extract multi-level skeleton features while Multi-View (MV) mechanism encourages different IGNet models to generate complementary features. Separable parametric graph convolution (SPG-Conv) is designed to enhance spatial modeling capability while reducing memory and computational cost. We also design a simple yet effective global context adaption (GCA) module to automatically "select" proper input-related interactions. On three challenging benchmarks, the proposed MV-IGNet shows impressive performance and much higher efficiency than other mainstream works.

ACKNOWLEDGMENTS

This work was supported by National Key Research and Development Program of China (2016YFB1001003), National Science Foundation of China (U20B200011, U1611461, U19B2035, 61527804, 61976137), STCSM (18DZ1112300). The authors would like to acknowledge the (partial) support from the Open Project Program (No.KBDat1604) of Jiangsu Key Laboratory of Big Data Analysis Technology, Nanjing University of Information Science & Technology.

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